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PDF Information Extraction, NDS Event Matching & NLP Analysis

eiLink RA Data Science Technical Task

A review-ready pipeline that turns drilling report PDFs into structured data, matched source events, and searchable NLP outputs.

From unstructured reports to usable intelligence

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OVERVIEW

Drilling daily reports are operational memory: what happened, when, where, and with which equipment. The problem is that this memory is trapped in PDFs and technical free text.

1

Extract

Parse drilling daily reports into report-level metadata, operation rows, summaries, fluids, failures, and diagnostics.

2

Match

Connect each NDS event to the most likely operation entry and source PDF using text similarity and domain evidence.

3

Analyze

Apply NLP to remarks and summaries to extract entities, classify activities, and surface TF-IDF keywords.

Reviewer takeaway

The project is easiest to follow as one pipeline with three outputs: a structured database, matched-event Excel workbook, and NLP outputs.

What is inside a drilling daily report?

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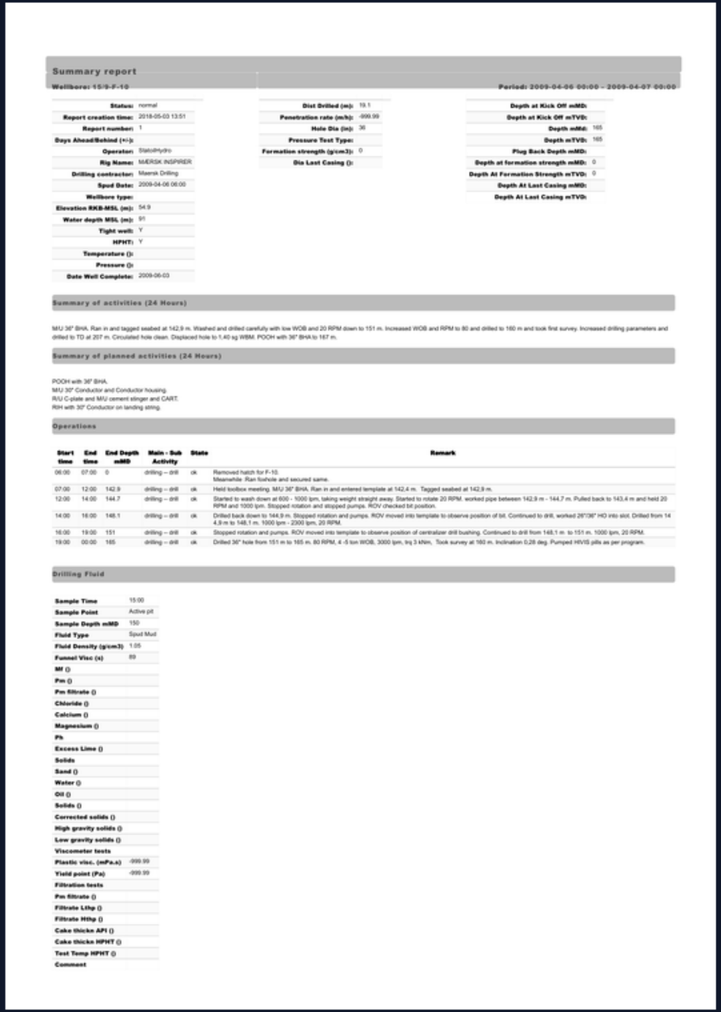
DOMAIN BACKGROUND



Summary report

Wellbore: 15/9-F-10 — Period: 2009-04-06 00:00 - 2009-04-07 00:00

Status:	normal
Report creation time:	2018-05-03 13:51
Report number:	1
Days Ahead/Behind (+/-):	
Operator:	StatoilHydro
Rig Name:	MÆRSK INSPIRER
Drilling contractor:	Maersk Drilling
Spud Date:	2009-04-06 06:00
Wellbore type:	
Elevation RKB-MSL (m):	54.9
Water depth MSL (m):	91
Tight well:	Y
HPHT:	Y
Temperature ():	
Pressure ():	
Date Well Complete:	2009-06-03

Dist Drilled (m):	19.1
Penetration rate (m/h):	-999.99
Hole Dia (in):	36
Pressure Test Type:	
Formation strength (g/cm3):	0
Dia Last Casing ():	

Depth at Kick Off mMD:	
Depth at Kick Off mTVD:	
Depth mMD:	165
Depth mTVD:	165
Plug Back Depth mMD:	
Depth at formation strength mMD:	0
Depth At Formation Strength mTVD:	0
Depth At Last Casing mMD:	
Depth At Last Casing mTVD:	

Summary of activities (24 Hours)

M/U 36" BHA. Ran in and tagged seabed at 142,9 m. Washed and drilled carefully with low WOB and 20 RPM down to 151 m. Increased WOB and RPM to 80 and drilled to 160 m and took first survey. Increased drilling parameters and drilled to TD at 207 m. Circulated hole clean. Displaced hole to 1,40 sg WBM. POOH with 36" BHA to 167 m.

Summary of planned activities (24 Hours)

POOH with 36" BHA.
M/U 30" Conductor and Conductor housing.
R/U C-plate and M/U cement stinger and CART.
RIH with 30" Conductor on landing string.

Operations

Start time	End time	End Depth mMD	Main - Sub Activity	State	Remark
06:00	07:00	0	drilling -- drill	ok	Removed hatch for F-10. Meanwhile: Ran foxhole and secured same.
07:00	12:00	142.9	drilling -- drill	ok	Held toolbox meeting. M/U 36" BHA. Ran in and entered template at 142,4 m. Tagged seabed at 142,9 m.
12:00	14:00	144.7	drilling -- drill	ok	Started to wash down at 600 - 1000 lpm, taking weight straight away. Started to rotate 20 RPM. worked pipe between 142,9 m - 144,7 m. Pulled back to 143,4 m and held 20 RPM and 1000 lpm. Stopped rotation and stopped pumps. ROV checked bit position.
14:00	16:00	148.1	drilling -- drill	ok	Drilled back down to 144,9 m. Stopped rotation and pumps. ROV moved into template to observe position of bit. Continued to drill, worked 26"/36" HO into slot. Drilled from 144,9 m to 148,1 m. 1000 lpm - 2300 lpm, 20 RPM.
16:00	19:00	151	drilling -- drill	ok	Stopped rotation and pumps. ROV moved into template to observe position of centralizer drill bushing. Continued to drill from 148,1 m to 151 m. 1000 lpm, 20 RPM.
19:00	00:00	165	drilling -- drill	ok	Drilled 36" hole from 151 m to 165 m. 80 RPM, 4 -5 ton WOB, 3000 lpm, trq 3 kNm, Took survey at 160 m. Inclination 0,28 deg. Pumped HIVIS pills as per program.

Drilling Fluid

Sample Time	15:00
Sample Point	Active pit
Sample Depth mMD	150
Fluid Type	Spud Mud
Fluid Density (g/cm3)	1.05
Funnel Visc (s)	89
Mf ()	
Pm ()	
Pm filtrate ()	
Chloride ()	
Calcium ()	
Magnesium ()	
Ph	
Excess Lime ()	
Solids	
Sand ()	
Water ()	
Oil ()	
Solids ()	
Corrected solids ()	
High gravity solids ()	
Low gravity solids ()	
Viscometer tests	
Plastic visc. (mPa.s)	-999.99
Yield point (Pa)	-999.99
Filtration tests	
Pm filtrate ()	
Filtrate Lthp ()	
Filtrate Hthp ()	
Cake thickn API ()	
Cake thickn HPHT ()	
Test Temp HPHT ()	
Comment	

Header / metadata

Well ID, report number, reporting period, status, operator, rig, contractor, depth flags.

Operations table

Time-based activity rows: start/end time, end depth, activity, state, and remark text.

Remarks

The most valuable free text for matching and NLP because it describes real operational events.

Tables & summaries

Fluid measurements, failure records, 24h summaries, and planned activities.

What the assignment asks for

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TASK OVERVIEW

The task is not only PDF parsing. It requires a connected workflow: extraction, database storage, event matching, benchmarking, and NLP analysis.

Task 1: PDF Parsing & Structured Extraction

- Header metadata
- Operation rows
- Equipment failures
- Summaries + fluids
- Other sections + DB

Task 2: NDS Event Matching

- Read nds_events.xlsx
- Search operation candidates
- Score text similarity
- Return PDF + excerpt
- Benchmark methods

Task 3: NLP Analysis

- Named entities
- Activity classification
- TF-IDF keywords
- CSV/SQLite outputs
- Quality validation

End-to-end solution architecture

PIPELINE

Each stage has a clear responsibility. This makes the work easier to explain, debug, and reproduce.

Input
PDF_VERSION_1000
nds_events.xlsx

Extraction
parse tables
detect sections

Database
SQLite tables
report_id links

Matching
NDS event →
operation entry

NLP
entities
tags
keywords

Outputs
Excel
CSV
SQLite
slides

Design principle

Every extracted row remains traceable to the source PDF, while the original text is preserved for matching and NLP.

Task 1: what had to be extracted

TASK 1

Header metadata

Well ID, period, status, operator, rig, contractor, flags

Operations

start/end time, end depth, activity, state, remarks

Daily Report PDF

Depths & tests

mMD/mTVD values, formation strength, hole diameter, pressure tests

Failures / fluids

equipment downtime, density, viscosity, yield point, sample depth

The important point: preserve structured fields and original narrative text.

Task 1 implementation: extraction workflow

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TASK 1

1. Read PDF

Load report file and capture raw text/tables for traceability.

2. Detect sections

Identify header, operations, summaries, fluids, failures, and extra sections.

3. Normalize fields

Convert repeated report sections into consistent rows and columns.

4. Save SQLite

Write relational tables and log extraction errors separately.

Preserve raw text supports audits

Preserve remarks used in Task 2 + NLP

Separate sections clean tables

Log errors quality control

Task 1 database schema: simple view

TASK 1

One PDF report becomes one row in reports. Repeated sections become child tables linked through report_id.

Full schema from dbdiagram.io

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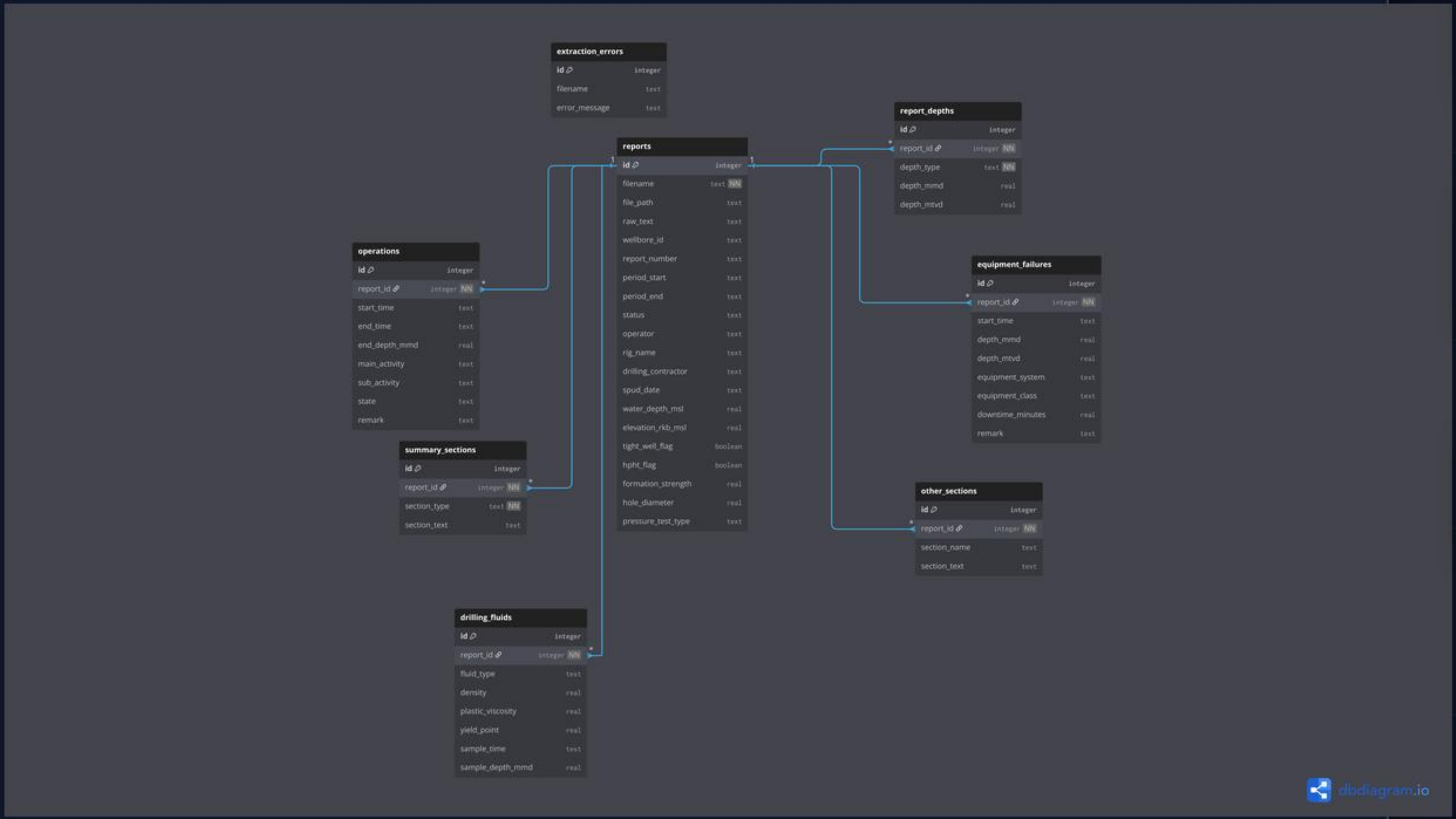
TASK 1



How to read it

reports is the parent table. Extracted sections connect back using report_id.

Why it matters

Every row remains traceable to a source PDF while repeated sections stay clean.

What to explain

Focus on table relationships, not every individual column.

Schema source: uploaded dbdiagram.io PDF

Why the schema is useful for the next tasks

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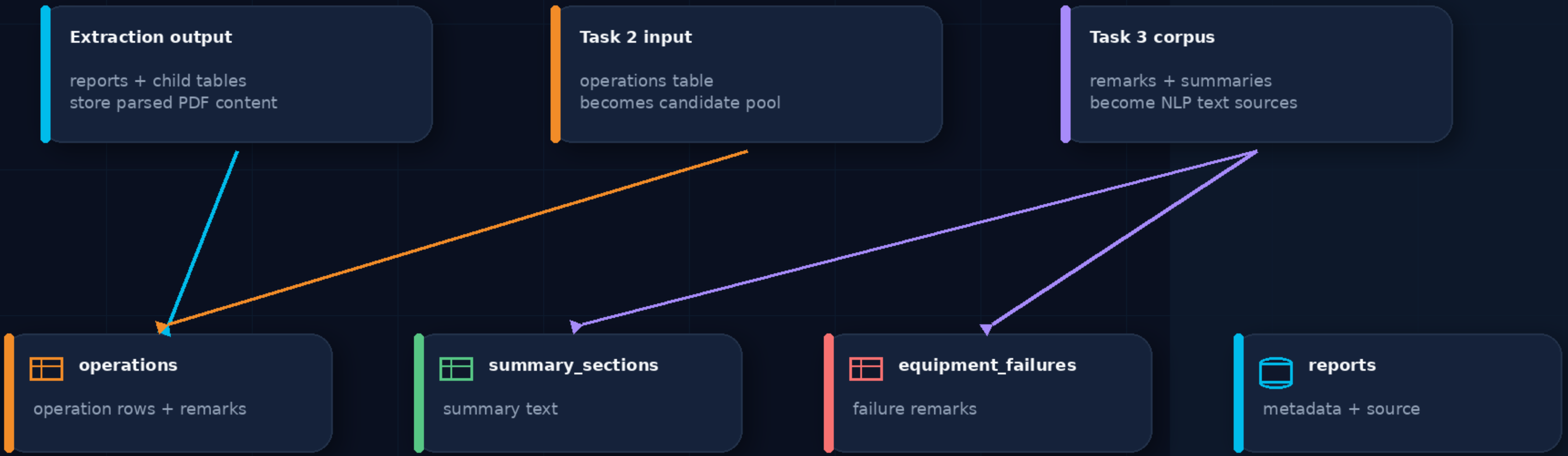
TASK 1



Smart design choice

The schema was built for reuse: extraction is not the end — it creates the structured foundation for matching and NLP.

Task 2: NDS event matching problem

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TASK 2

This is a source-tracing problem: each NDS event must be connected to the operation entry and PDF report that most likely produced it.

nds_events.xlsx

Well	Event text
15/9-F-10	inclination angle was higher than expected...
15/9-F-11	tight hole event while drilling interval...
15/9-F-12	excessive clay accumulation in BHA...
15/9-F-13	differential stuck while RIH with casing...

Candidate pool

Operation rows extracted from the SQLite database are filtered by well context and compared against the event text.

Required output

Matched PDF document
Similarity score
Matched textual excerpt
Benchmark sheets for the methods

Task 2 implementation: matching workflow

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TASK 2

1. **Event row**
   Well + Event text
2. **Candidate ops**
   operations table rows
3. **Normalize**
   clean casing, punctuation, tokens
4. **Score**
   TF-IDF + fuzzy + overlap + domain signals
5. **Best match**
   PDF + score + excerpt + top candidates

Why this is stronger than keyword search

It stores the final match, the top candidate alternatives, and method-level scores, making the result auditable instead of black-box.

Task 2 scoring: combine general similarity with drilling evidence

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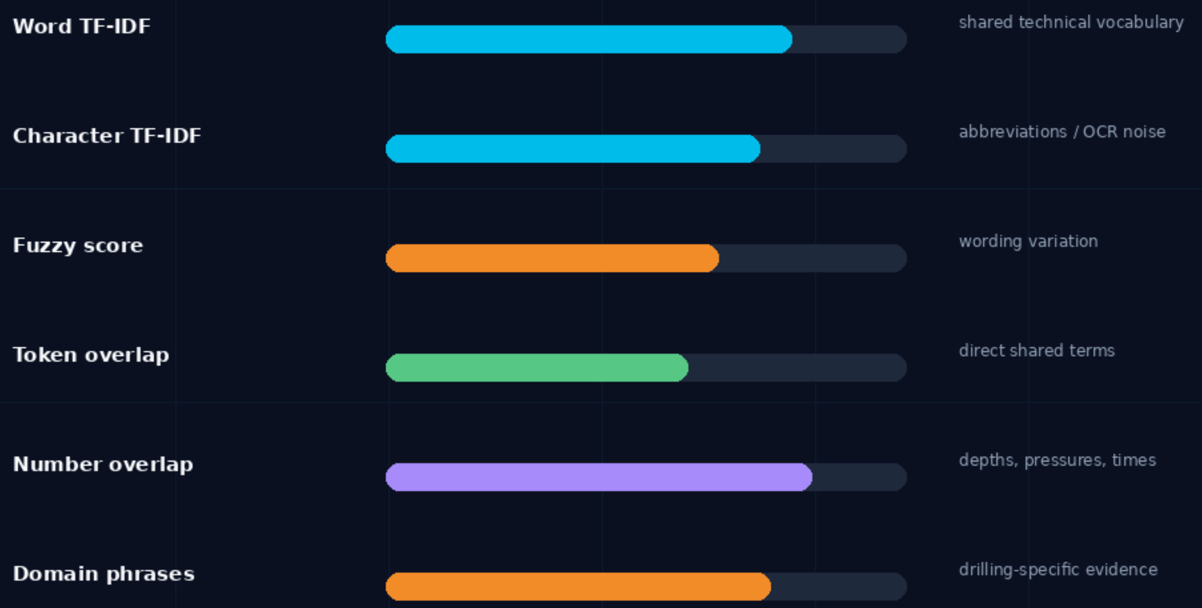
TASK 2

Similarity signals



Final score

weighted evidence from multiple methods

Key idea: measurements and drilling phrases prevent generic text similarity from dominating technical event matching.

Task 2 outputs: designed to be auditable

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TASK 2

task2_nds_event_matches.xlsx

best_matches	final selected match
top_candidates	audit trail of alternatives
method_benchmark	scores by method
method_summary	method comparison
score_summary	confidence distribution
candidate_counts	candidate availability
original_nds_events	raw input reference

The final answer is not enough

The reviewer should be able to see why a match was chosen and what alternatives were considered.

Top candidates make it reviewable

If a best match seems questionable, the candidate sheet makes the decision inspectable.

Benchmark sheets support the method

Method summaries show whether the scoring approach behaves consistently.

Task 3: NLP analysis pipeline

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TASK 3

operations.remark
operation free text

summary_sections
daily/planned activities

equipment_failures.remark
failure descriptions

NLP processing
entity rules
activity tags
TF-IDF vectorization

Entities
depths, equipment, measurements, times

Activity tags
TRIP_IN, CEMENT, WAIT, REPAIR...

Keywords
distinctive report-level TF-IDF terms

Purpose
Task 3 makes the database searchable by technical entities and normalized operational labels, not only raw text.

Task 3: entity extraction example

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TASK 3

Example operation remark

RIH with 20" casing. Differential stuck observed. Pumped 300 m3 seawater. Set down weight increased and string came free.

- **20" casing** — Equipment / size
- **Differential stuck** — Operational event
- **300 m3** — Measurement
- **seawater** — Fluid
- **set down weight** — Operational parameter

Extracted entity categories

Depths, equipment names, measurements, pressures, flow rates, weights, and time references.

Depth mMD / mTVD	**Equipment** BOP, TDS, BHA
Pressure bar / psi	**Flow** lpm / m3
Weight MT / ton	**Time** hours / dates

Rule-based NER is practical because drilling language uses repeated units, abbreviations, and technical patterns.

Task 3: activity tags and report-specific keywords

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TASK 3

Activity classification

- TRIP_IN
- TRIP_OUT
- CUT
- CEMENT
- PRESSURE_TEST
- EQUIPMENT_FAILURE
- REPAIR
- WAIT
- CIRCULATE
- FISHING

TF-IDF keyword extraction

- tight hole
- BHA
- casing
- cement
- pressure test
- seawater
- drilling
- circulate

Why these NLP outputs matter

Activity tags normalize inconsistent human wording. TF-IDF keywords show what makes each report distinctive and help identify unusual problems.

Quality control: fix, rerun, validate

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VALIDATION

Issue found

Task 3 TF-IDF token pattern created a Python regex range error.

Fix applied

Move hyphen to safe position in the character class.

Validation

Restart runtime and run all notebooks from top to bottom.

Outputs

Updated SQLite, CSV files, NLP zip, matched events Excel, slides.

Corrected TF-IDF token pattern

```
token_pattern = r"(?u)\b[a-zA-Z][a-zA-Z0-9_/-]{1,}\b"
```

The hyphen is placed safely at the end of the character class, so it is not interpreted as an invalid range.

Final submission package and reviewer path

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SUBMISSION

The final package should make it easy for reviewers to reproduce the pipeline and inspect each output.

Task 1 notebook
PDF → SQLite extraction

Task 2 notebook
NDS matching workflow

Task 3 notebook
NLP analysis

well_reports.sqlite
structured database

Task 2 Excel
best matches + benchmarks

Task 3 zip/CSV
entities + tags + keywords

Datasets
PDFs + nds_events.xlsx

Presentation
methodology summary

Reviewer path

Run notebooks
verify reproducibility

Open DB
inspect schema and row counts

Open Excel
review matches and benchmarks

Open NLP outputs
inspect entities/tags